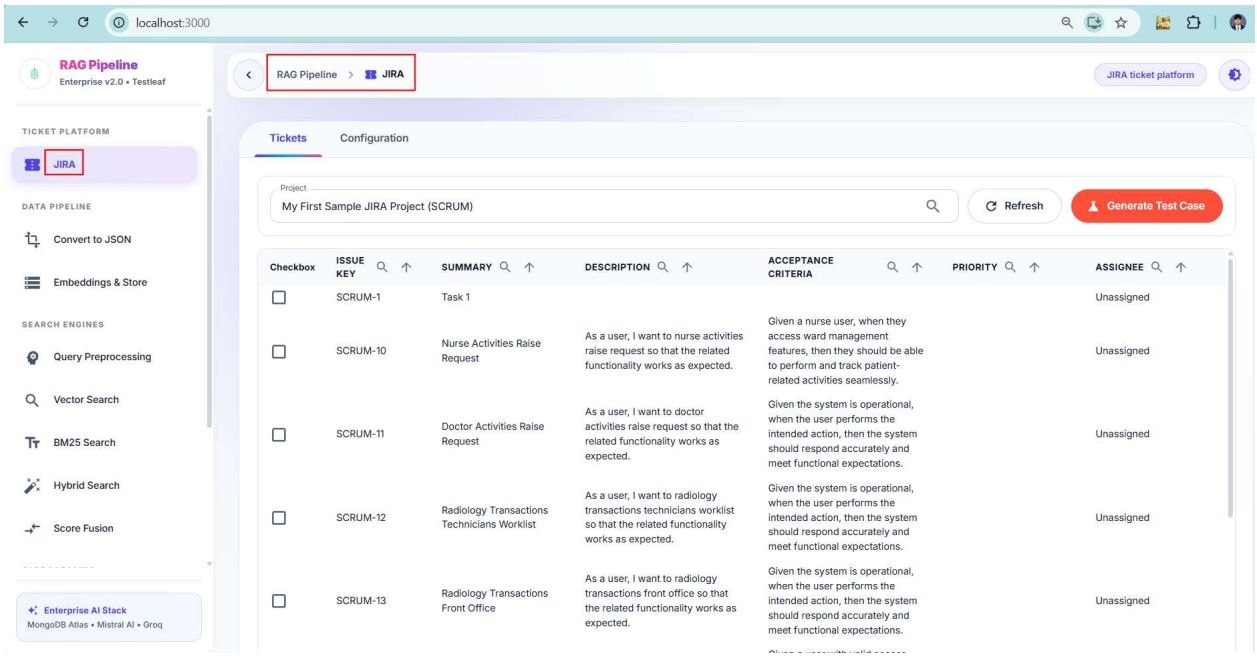
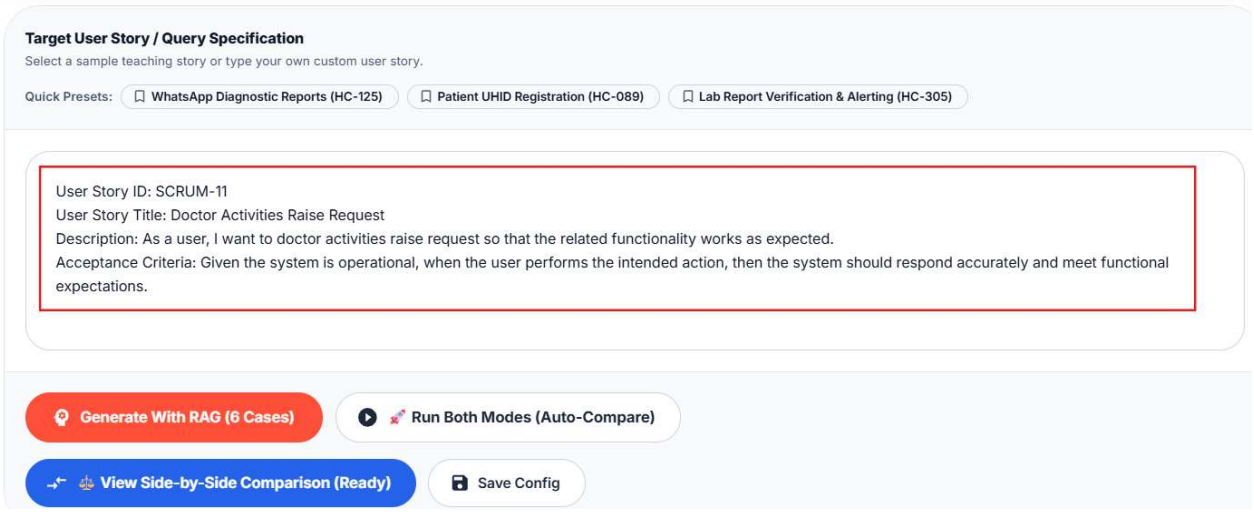
Home Assignment - JIRA integration to RAG pipeline APP







RAG vs Non-RAG Generated Test Cases Comparison

Contrasting **Generated Test Cases (Without RAG)** against **Generated Test Cases (With RAG)** for the exact same user story.

⚠ Comparison Requires Both Pipelines to Be Generated

To compare the effect of RAG augmentation, please run generation for both **Without RAG** (baseline) and **With RAG** (10-step pipeline).

1. Without RAG (Baseline) Generated

✅ 3 baseline test cases generated directly by LLM.

2. With RAG (Pipeline) Not Run Yet

Complete 10-step MongoDB retrieval pipeline has not been executed yet.

Generate With RAG Now

Generate Both Pipelines (Auto-Compare)

Executive Dimension Comparison Matrix

5 fundamental architectural differences between Zero-Context baseline generation and RAG retrieval augmentation.

DOMAIN GROUNDING

Base: Low (Generic)

RAG: High (UHID/PRN)

RAG grounds tests in hospital entities, barcodes, and HIPAA standards.

DATABASE GROUNDING

0 vs **8 Cases**

Searches 6,000+ MongoDB cases via Hybrid Search (BM25 + Vector).

STEP DEPTH & DETAIL

Base: 3-4 Generic Steps

RAG: 5-8 Detailed Steps

RAG enforces field-level verification in database tables.

HALLUCINATION RISK

Base: Elevated

RAG: Minimal

RAG verifies against real existing hospital test suites.

COST & TOKENS

Base: 2810t ($0.000225)

RAG: 4169t ($0.000334)

Modest token difference for vastly superior quality.